CHAPTER 8

Iteration and recursion

Learning Objectives

After learning the concepts in this chapter, the students will be able

- To know the concepts of variants and invariants used in algorithmic techniques.
- Apply algorithmic techniques in iteration and recursion process.

There are several problems which can be solved by doing the same action repeatedly. Both iteration and recursion are algorithm design techniques to execute the same action repeatedly. What is the use of repeating the same action again and again? Even though the action is the same, the state in which the action is executed is not the same. Each time we execute the action, the state changes. Therefore, the same action is repeatedly executed, but in different states. The state changes in such a way that the process progresses to achieve the desired input-output relation.

Iteration: In iteration, the loop body is repeatedly executed as long as the loop condition is true. Each time the loop body is executed, the variables are updated. However, there is also a property of the variables which remains unchanged by the execution of the loop body. This unchanging property is called the loop invariant. Loop invariant is the key to construct and to reason about iterative algorithms.

Recursion: Recursion is another algorithm design technique, closely related to iteration, but more powerful.Using recursion, we solve a problem with a given input, by solving the same problem with a part of the input and constructing a solution to the original problem from the solution to the partial input.

DO YOU KNOW?

E W Dijkstra was one of the most influential pioneers of Computing Science. He made fundamental contributions in diverse areas such as programming language design, operating systems and program design. He coined the phrase "structured programming" which helped lay the foundations for the discipline of software engineering. In 1972, he was awarded ACM Turing Award, considered the highest distinction in computer science. Dijkstra is attributed to have said "Computer science is no more about computers than astronomy is about telescopes."



8.1 Invariants

Example 8.1. Suppose the following assignment is executed with (u, v) = (20,15). We can annotate before and after the assignment.

-- before: u, v = 20, 15

u, v :=u+5,v-5

-- after: u, v = 25, 10

After assignment (u, v) = (25, 10). But what do you observe about the value of the function u + v?

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